- A single convex lens can act as a simple microscope when object is within the focal length.
- In near point focusing, the image is formed at $D = 25$ cm.
- The magnification in near point focusing is, $m = 1 + \frac{D}{f}$
- In normal focusing, the image is formed at infinity, the magnification in normal focusing is, $m = \frac{D}{f}$
- The resolving power of microscope is, $R_{\mathrm{M}} = \frac{1}{d_{\min}} = \frac{2(NA)}{1.22\lambda}$
- The resolving power of telescope is, $R_T = \frac{1}{r_0} = \frac{a}{1.22\lambda f}$
- Magnification in near point focusing in compound microscope is, $m = m_o m_e = \left(\frac{L}{f_o}\right)\left(1 + \frac{D}{f_e}\right)$
- Magnification in normal focusing in compound microscope is, $m = m_o m_e = \left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right)$
- Magnification in astronomical telescope is, $m = \frac{f_o}{f_e}$ and the tube length is, $L = f_o + f_e$
- The correction lens for nearsightedness/myopic eye should be a concave lens with focal length, $f = -x$. Where x is the maximum distance upto which an object can be seen.
- The correction lens for farsightedness/hypermetropic eye should be a convex lens with focal length, $f = \frac{y \times 25\ \mathrm{cm}}{y - 25\ \mathrm{cm}}$. Where, the 25 cm is the near distance and y is the minimum distance upto which an object can be seen.
- The farsightedness arising due to aging is called *presbyopia*.
- Astigmatism is the defect arising due to different curvatures along different planes in the *eye* lens.

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